Chase Golden – Assignment #3

Segmentation and Profiling

With this project assessment, I investigated methods of advanced customer segmentation through the use of a customer dataset that contained a number of demographic, behavioral, and financial attributes about customers. The business problem was simply to identify customer heterogeneity for the company to better develop customer-specific marketing strategies, to improve the personalization of customer service, and to reduce customer churn. To develop these customer segments, I applied and compared two distinct methodologies: K-Means clustering and rules-based segmentations based on business logic. This approach supports an analysis that incorporates a clearly rigorous statistical approach to provide insights into customer patterns and a rules-based segmentation that provides an interpretable segmentation that is consistent with how the stakeholders understand their business. The ultimate goal of all this is to identify actionable insights, or customer-specific improvements to their business.

1 Segmentation Approach

Two segmentation approaches were utilized. The first method was K-Means clustering, which is a type of algorithm that identifies patterns automatically by clustering customers with similar characteristics across multiple numeric variables such as age, household size, financial flexibility, technology engagement, monthly data consumption, and number of premium subscriptions. K-Means identifies clusters at the point where you stop seeing benefits by merging data. Thus, the clusters that emerge from data when applying K-Means provide an unbiased advantage, as you can plan for emergent behavioral and demographic groupings that you may not have wanted or even considered as a measurable group.

The second approach to customer segmentation was rules-based segmentation that was created using business logic. This means building segments of customers based on engineered categorical variables, like streaming intensity, level of credit card usage, and level of debt stress. Rules-driven segmentation is designed with emphasis on intelligibility, which makes it easier in principle for stakeholders to consider customer groupings and to use their observations on customer groupings modeled based on a threshold they deem helpful or fair.
In conclusion, the two methods present a mix of the technical aspects of machine learning and the fusion of business logic on a domain-based categorization classification.

KMeans Segment Card

Cluster KMeans 4	Avg. Age	Avg. Financial Flexibility Score	Avg. Household size	Avg. Monthly data usage	Avg. Premium Subscription Count
Cluster 1	44.80	47.88	2.28	35.03	2.60
Cluster 2	41.65	50.10	2.19	26.24	1.77
Cluster 3	46.76	69.57	2.27	49.82	2.78
Cluster 4	45.35	69.02	2.30	49.98	3.08
Not Clustered	47.90	39.75	2.18		1.05

Measure Values

1.05 69.57

Figure 1: K-Means Segment Card

Rules Segment Card

Segment Rules	Avg. Age	Avg. Financial Flexibility Score	Avg. Household size	Avg. Monthly data usage	Avg. Premium Subscription Count
High-Value Digital	47.62	64.17	2.22	40.98	1.48
Medium-Value Omnicha..	47.45	22.62	2.45	32.10	1.48
Price-Sensitive / Low En..	46.83	39.09	2.19	39.44	1.41

Measure Values

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Figure 2: Rules Segment Card

2 Key Findings

The comparative analysis yielded several valuable insights. K-Means Cluster 4 was comprised of what could be described as the oldest and most financially flexible customers, who had high premium service usage and the highest average data usage. This was a cohort with significant potential for high-end product adoption and loyalty opportunities. K-Means Cluster 2 was the youngest and financially least flexible group. They were the only cohort without premium subscriptions and exhibited low-tech engagement scores, making them a low-value segment in the context of the current products offered. ... On the rules-based side, the "High-Value Digital" segment had higher levels of financial flexibility and data usage, but to a slightly lesser extent than the high-end K-Means Cluster 4 of customers described above. Honestly, I was surprised to see that the "Price-Sensitive" segment of customers appeared younger and financially constrained, but still exhibited surprisingly high levels of data usage, and that they appeared digital-friendly, despite their economic constraints. Observing the differences across segments revealed more in-depth value variables between customers that any vendor would miss if relying purely upon demographic profiling.

3 Recommendations

When thinking of short-term growth, opportunities exist mainly with the High-Value Digital (from the rules) and K-Means Cluster 4 segments. These customers are confident with technology and have disposable income, and are already using digital services, so it gives me a defined scope of potential. The business can use some kind of customized bundled offers, cross-selling strategies and loyalty efforts, etc.
For mid-tier opportunities, I may use K-Means Cluster 3 and the "Medium-value Omnichannel" group, which had moderate usage and engagement levels. These customers may respond better to services that focus on "convenience" or bundles where offering more perceived value could be linked to less upfront financial commitment.

Finally, the Price Sensitive & Cluster 2 customers who will require value-driven solutions could look at offering flexible plans that fit their budgets, or other educational content that increases platform usage over time.

4 Data Preparation & Feature Engineering

The original data set of over 5,000 customer records and greater than 60 columns was cleaned up by deleting personally identifiable information (PII) and non-informative variables. Null values were dealt with, and category standardization was completed. A number of new variables (e.g., Tech Engagement Score, number of Premium Subscriptions, Financial Flexibility Score) aimed to represent multi-variable behaviors (e.g., how frequently smart technology devices were used, how frequently payments were made, etc.), analyses that were multivariate in nature. The final data set was saved and organized to be analyzed in Tableau, which showed new features/variable changes to ensure transparency in the data cleaning process.

5 Clusters and Rules

By using Tableau's native clustering engine, I developed a scatter plot visual of the Tech Engagement Score and the Financial Flexibility Score on the axes. I added Monthly Data Usage and Premium Subscription Count as details, so that Tableau could recognize them using multiple continuous dimensions. The clustering model was set up to be four clusters because of interpretability, and also because they are visually separable. The cluster labels were renamed and converted to dimensions, which allowed crosstab analysis to be completed across other worksheets. The cluster results showed meaningful variation between age, financial metrics, and product engagement.

In order to perform segmentation according to business-defined criteria, I created a calculated field in Tableau based on if-then logic for:" streaming intensity, credit card usage level", and" debt stress level"; these categories were all derived in Tableau prep, and they were really important for the next piece of business logic. Customers were binned into High-Value Digital, Medium-Value Omnichannel, and Price-Sensitive cohort groups. The output provided a clear and transparent segmentation for an executive or non-technical audience, all tied to the original business intent. While it's a far coarser-grained strategy, it does provide a clear lens to visualize customer priorities and strategically see segmentation.

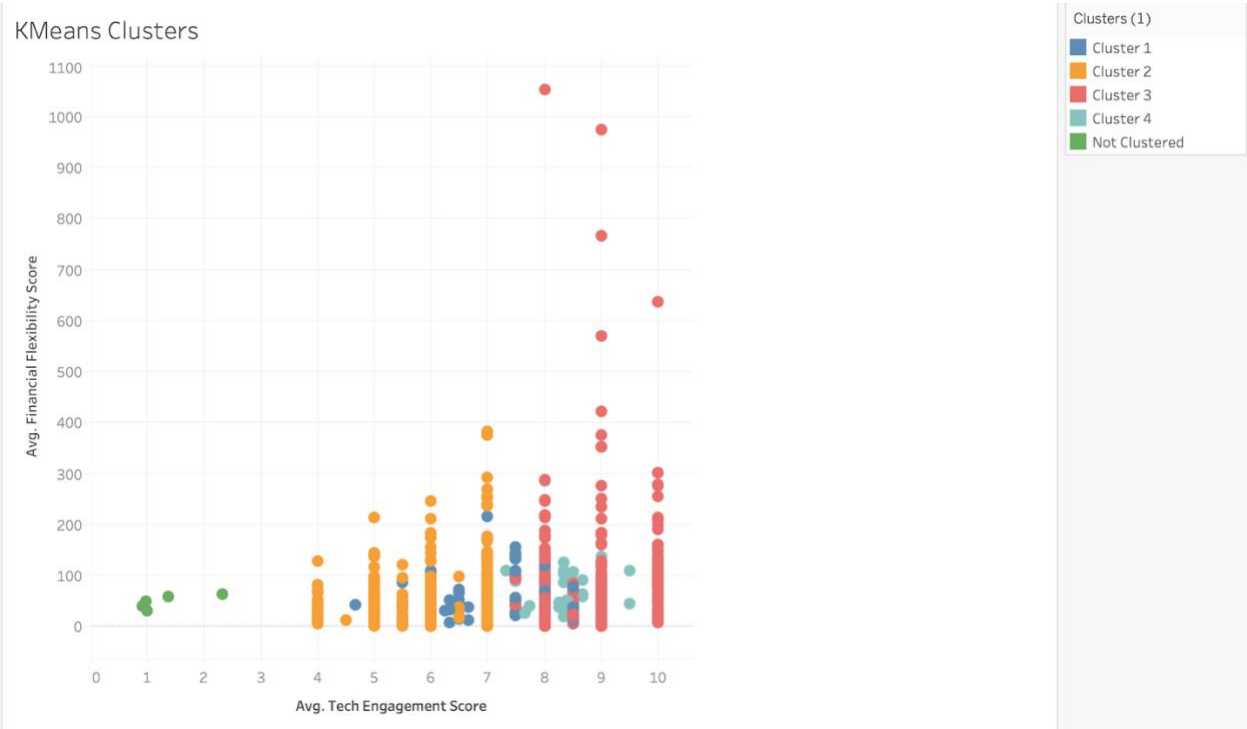

Figure 3: K-Means Clusters

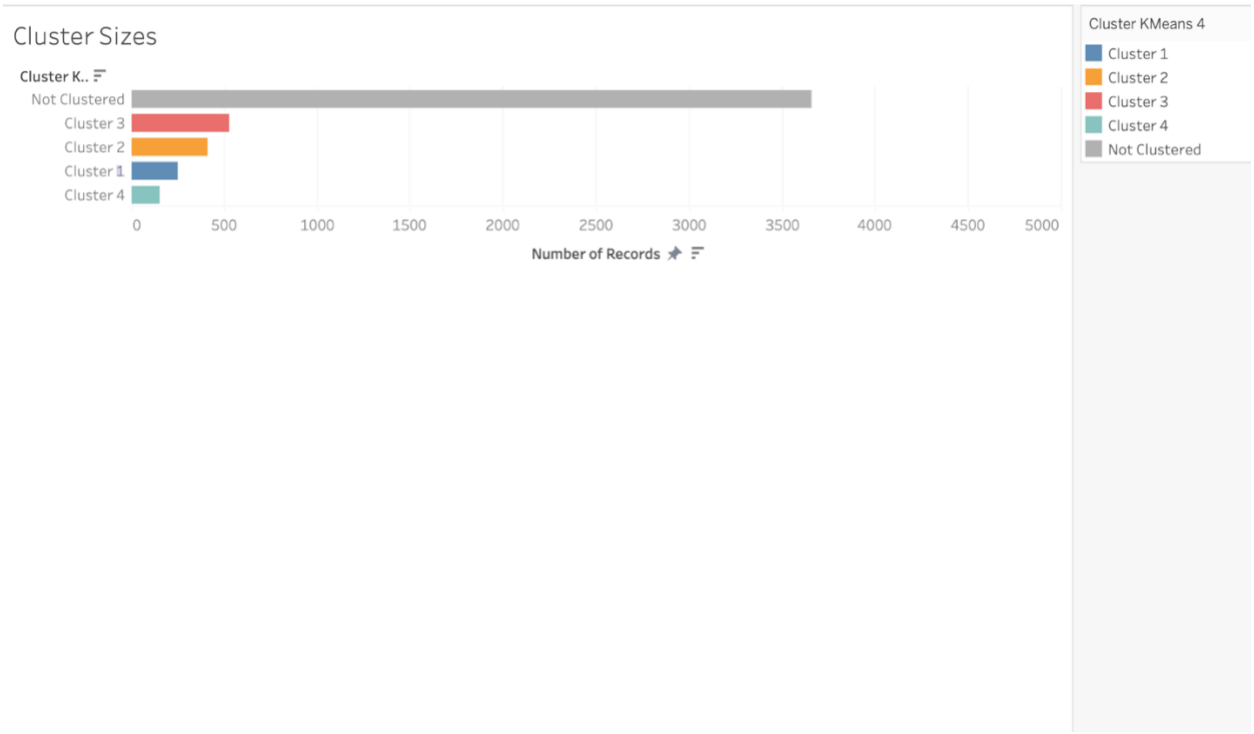

Figure 4: Cluster Sizes

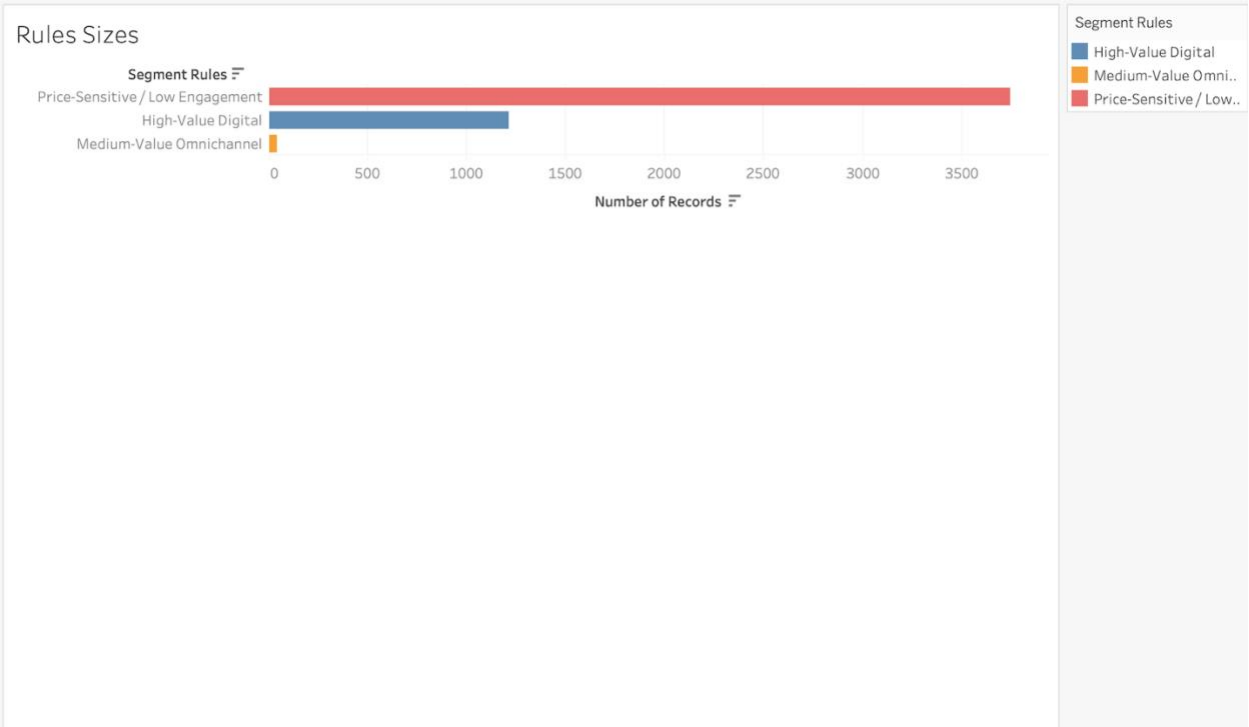

Figure 5: Rules Sizes

KMeans Profile

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Figure 6: K-Means Profile

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Measure Values

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Figure 7: Rules Profile

6 Comparing and Evaluating

Two profile tables were generated to consider segment differences on the measures of interest. The K-Means profile table presented the average age, data usage, financial flexibility, and so on. Cluster 4 dominated on most of these measures, and Cluster 2 had numbers that lagged behind. The rules profile table followed the same pattern, but there was more blending of segments. High-Value Digital customers had almost the same age and financial scores as Cluster 4, but had a slightly lower number of premium subscriptions. Looking at both profiles gives some verification of the utility and logic underlying both approaches to segmentation and links to business goals.

K-Means provided useful statistical separation and good insight into behavior clusters that delivered distinctions in nuanced ways. K-Means is a strong method for building predictive models and for lifecycle-based marketing. However, K-Means can sometimes be opaque for non-technical audiences. In contrast, the rules-based method provided less granularity than K-Means but was also easily explainable and intuitive in terms of campaign design. In the end, both methods should be used together; I will continue advocating for K-Means to be used in the backend to target based on predicted customer value and rules-based segmentation to help with strategic campaign planning with top executives. Combining both methods, I can allow the business to act on the customer value developed on the dataset from a computer-generated model and from a human-processed model (rules-based).

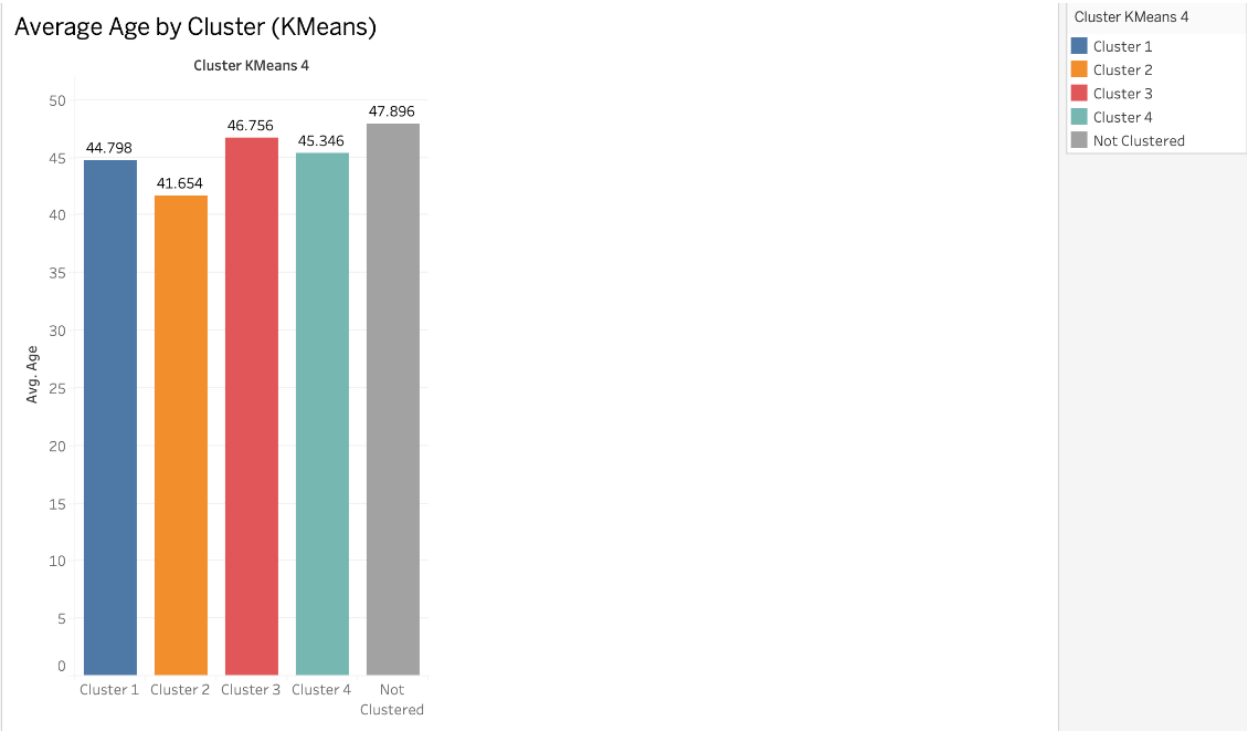

Figure 8: Age by Clusters

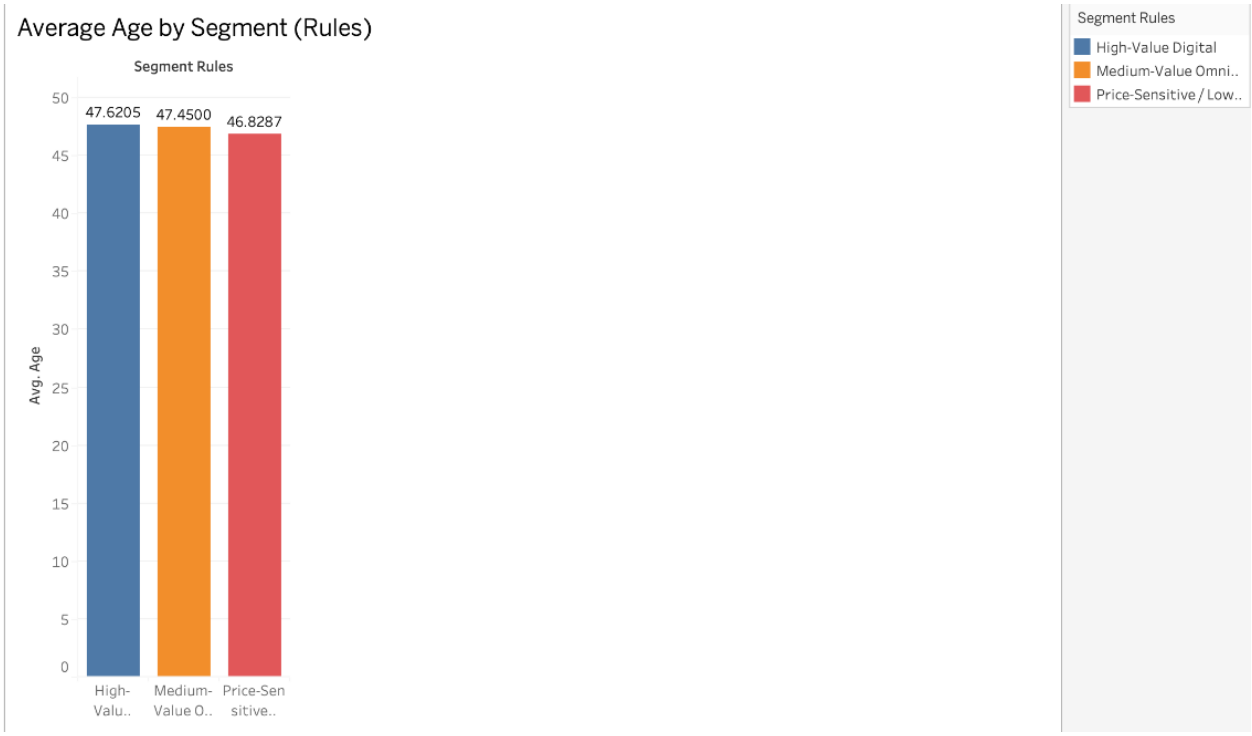

Figure 9: Age by Rules

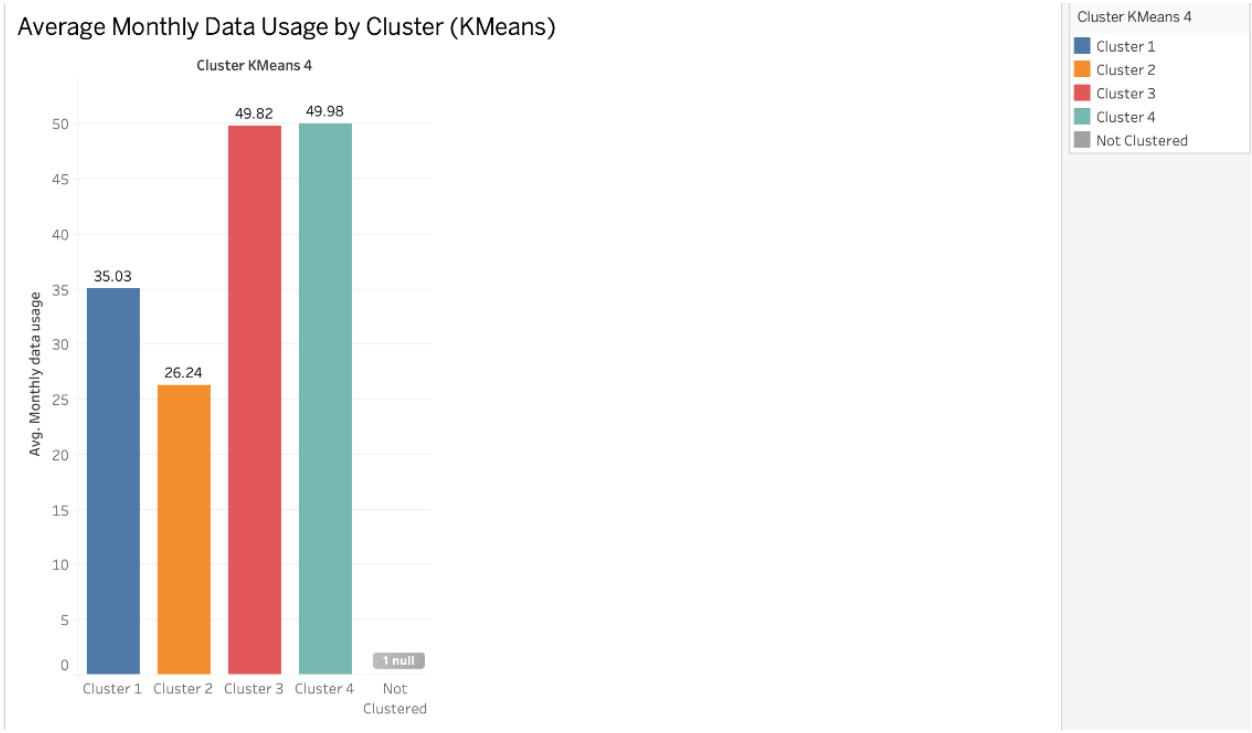

Figure 10: Monthly Data Usage by Clusters

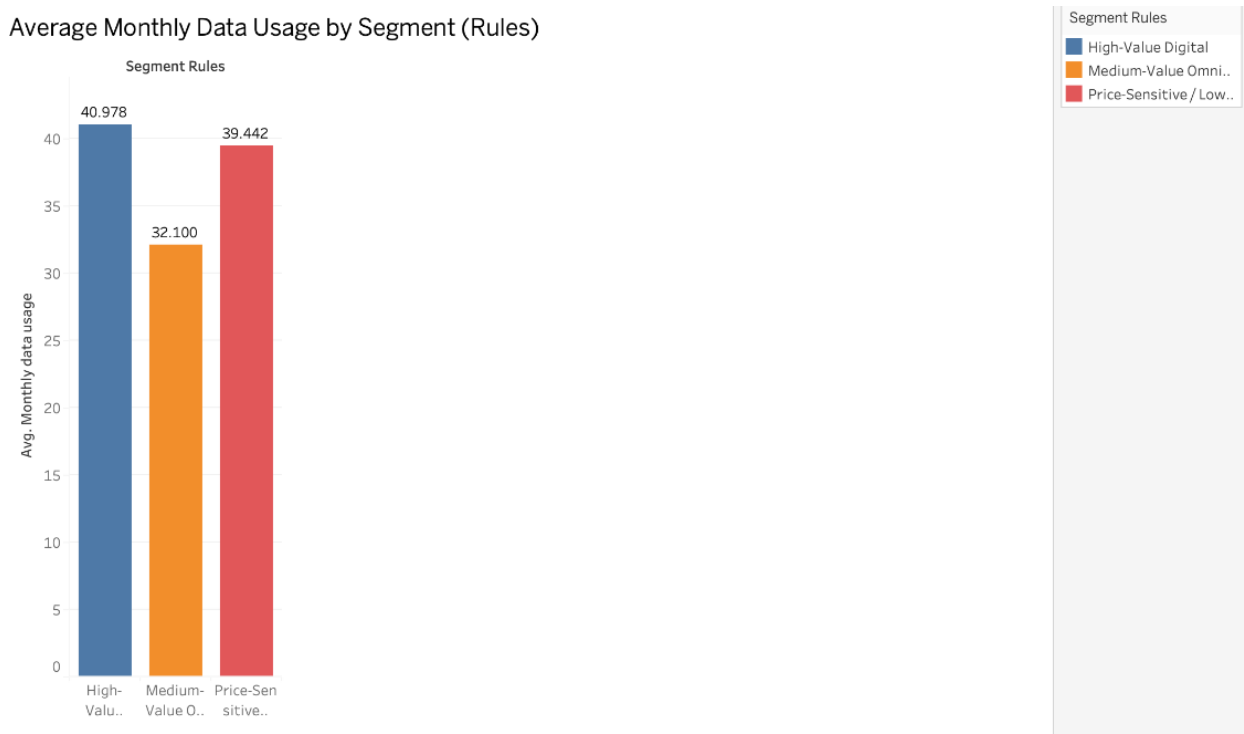

Figure 11: Monthly Data Usage by Rules

7 Conclusion

K-Means performance can be sensitive to scaling and outliers in the data, and while the 4 clusters are justifiable visually, they are still arbitrary. Rules-based logic can oversimplify yes/no outcomes and will miss mixed personas or subtle patterns in your data. However, all of these iterations could scaffold into hybrid modeling, testing campaign evidence of A/B testing segmented into social segments, taking behavioral time-series data into account. Segmentation should be a continuous and evolving process that aligns with your customer base.

The project successfully applied two different methods for customer segmentation and illustrated the advantages and drawbacks of each. The business now has two practical ways to understand and target different customer segments. The project further cemented best practices in feature engineering, Tableau modeling, and comparative assessments. The next step is to put these lessons into practice in a business setting: running campaigns, measuring KPIs by segment, and adjusting segmentation based on real-time feedback.

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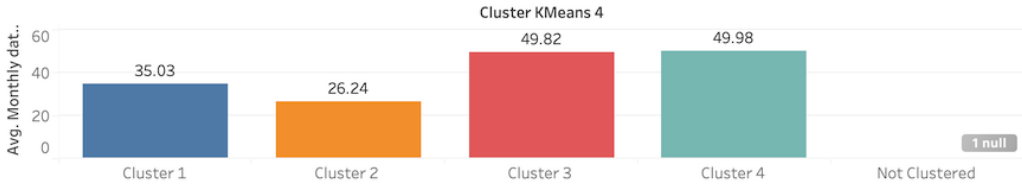
Measure Values

1.05 — 69.57

Cluster KMeans 4
- Cluster 1
- Cluster 2
- Cluster 3
- Cluster 4
- Not Clustered

Segment Rules
- High-Value Digital
- Medium-Value Omni..
- Price-Sensitive / Low..

Average Monthly Data Usage by Cluster (KMeans)



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Average Monthly Data Usage by Segment (Rules)

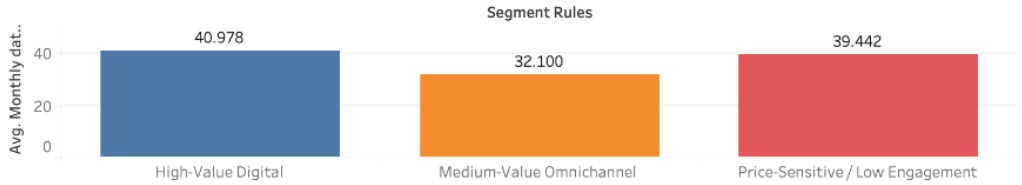

Figure 12: Segment Card and Monthly Data Usage Comparisons between Clusters and Rules

8 Appendix

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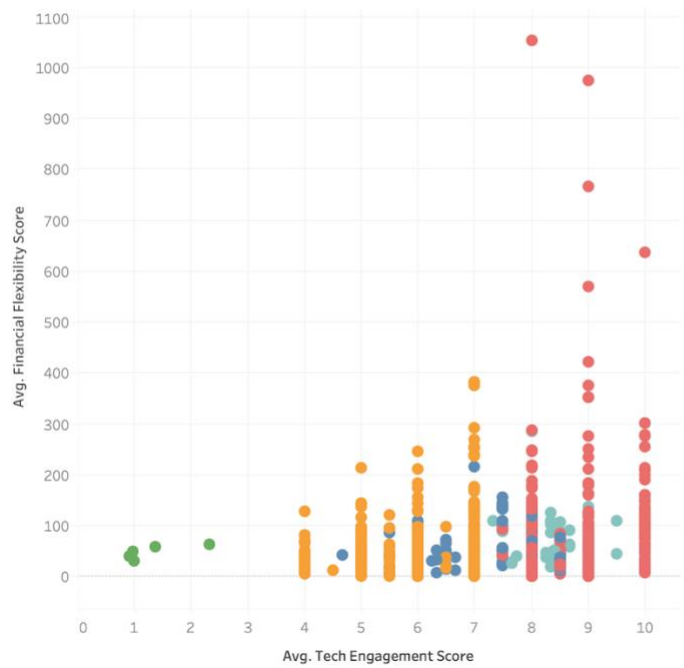
KMeans Clusters



Clusters (1)

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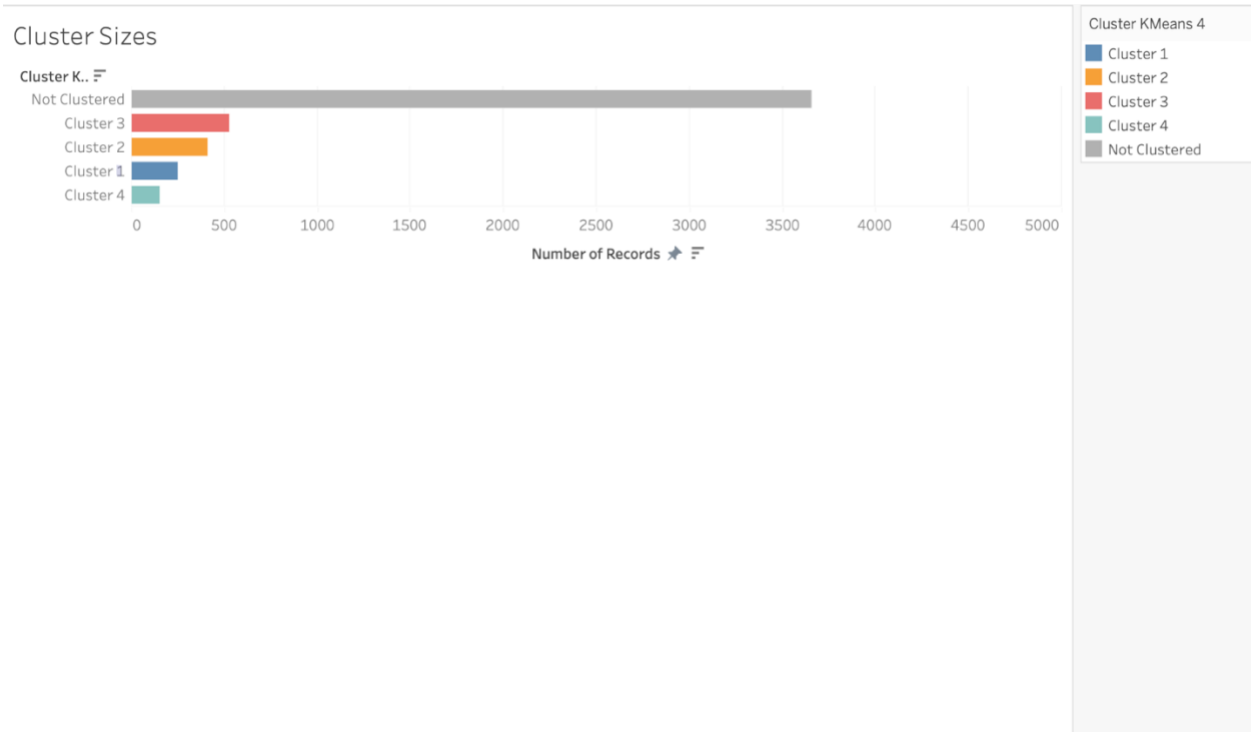
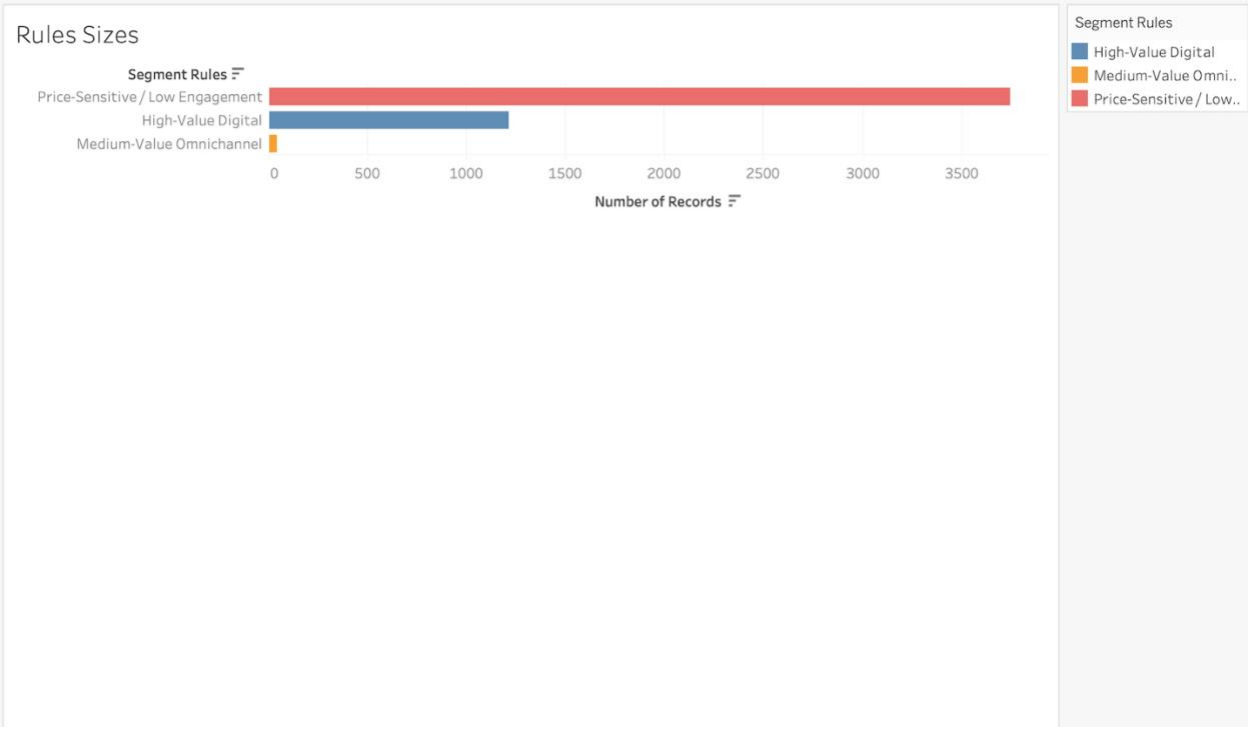

Cluster Sizes
Cluster K..
Not Clustered
Cluster 3
Cluster 2
Cluster 1
Cluster 4
0
500
1000
1500
2000
2500
3000
3500
4000
4500
5000
Number of Records
Cluster KMeans 4
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Not Clustered


Rules Sizes
Segment Rules
Price-Sensitive / Low Engagement
High-Value Digital
Medium-Value Omnichannel
0
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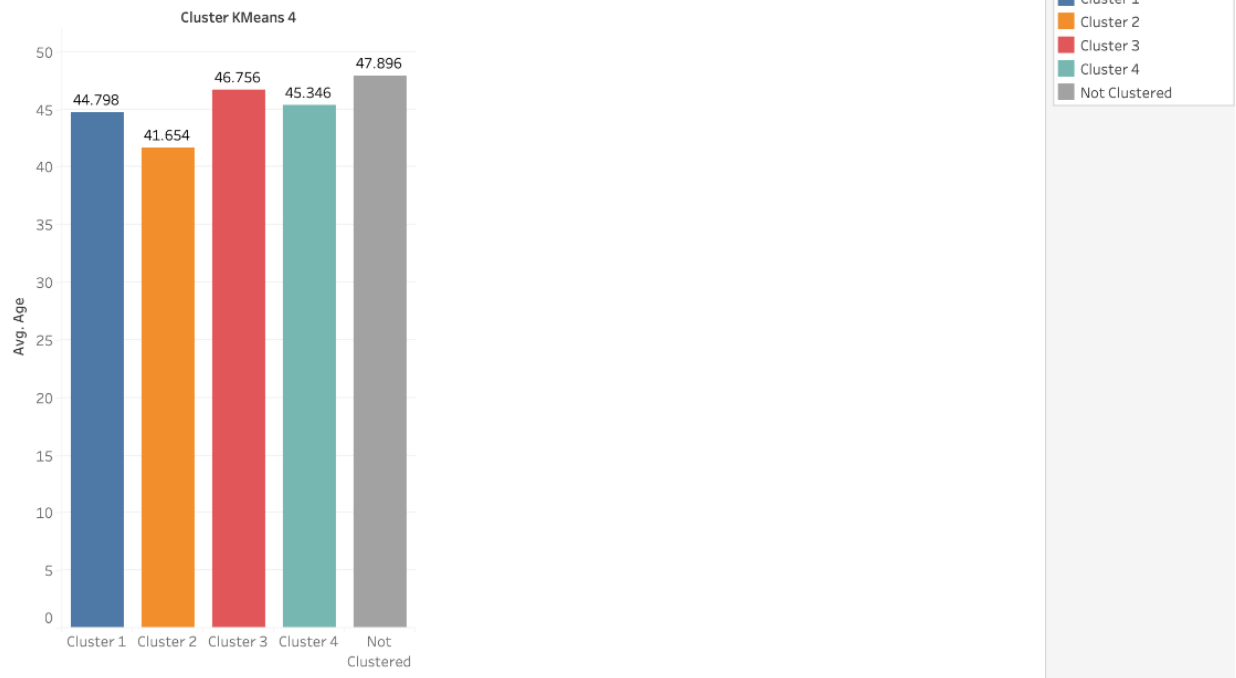
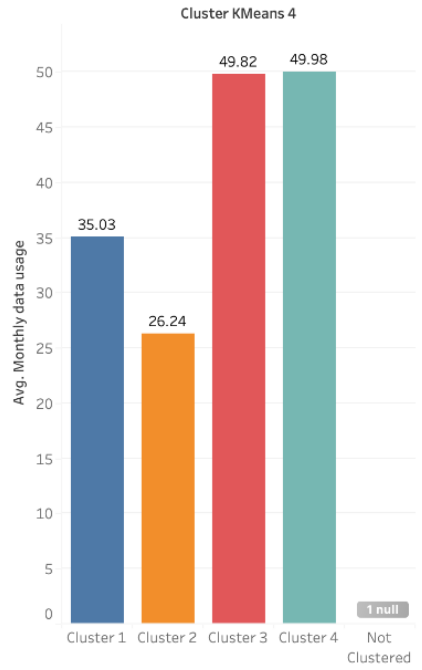
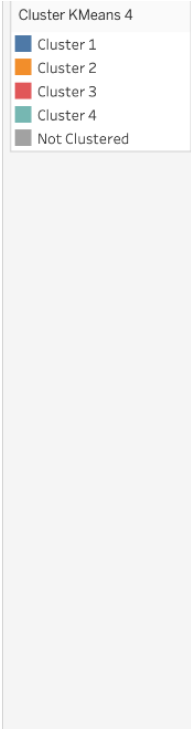
Average Age by Cluster (KMeans)



Average Age by Segment (Rules)

Average Monthly Data Usage by Cluster (KMeans)





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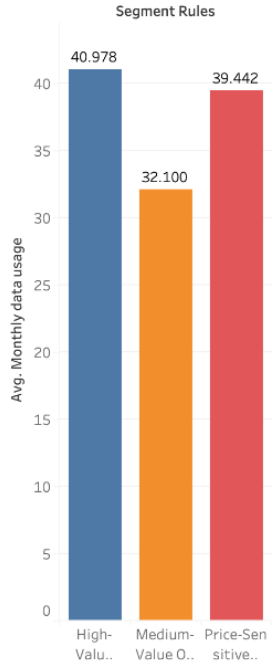

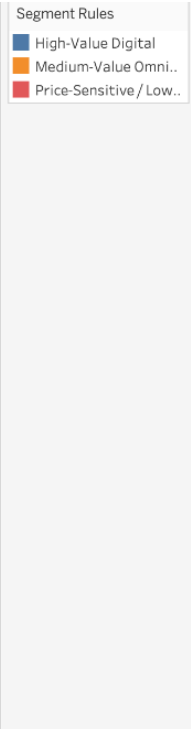

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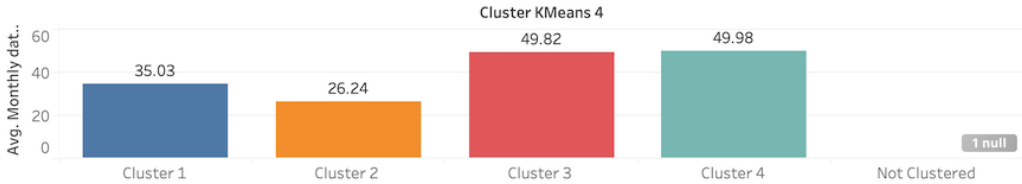

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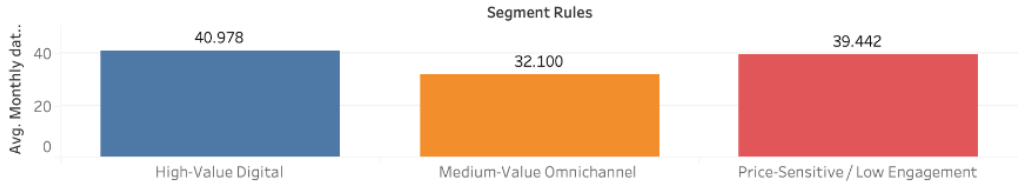
Average Monthly Data Usage by Segment (Rules)